Performance Stability (50 Users)

During a simulation with 50 concurrent users clicking buttons every two to four seconds, the calculator maintained perfect operation. The application successfully processed approximately 16 requests per second without any errors, delivering rapid response times of around 100 milliseconds for nearly every action.

Capacity Thresholds

Stress tests identified that the system can reliably support up to 2,000 simultaneous users. At this peak capacity, the server managed about 565 requests per second; however, exceeding this 2,000-user limit caused the server to become overloaded and begin dropping connections.

Addressing Latency in the "Add" Function

Feedback regarding intermittent slowness in the "Add" feature was validated by tests showing delays exceeding two seconds. Since these lags occur even under low usage, the issue is likely a code-level glitch that causes the system to freeze when processing specific number combinations.

Resolving Memory-Related Crashes

Reports of random application crashes were linked to the server reaching its 1.5GB memory limit. High request volumes trigger a "memory leak" in the code, which exhausts the available 1.5GB of memory and leads to a fatal system crash requiring a manual restart.